



IECEx Certificate of Conformity

INTERNATIONAL ELECTROTECHNICAL COMMISSION IEC Certification Scheme for Explosive Atmospheres

for rules and details of the IECEx Scheme visit www.iecex.com

Certificate No.: IECEx EPS 14.0094X

Issue No: 1

Certificate history:

Issue No. 1 (2018-02-28)

Issue No. 0 (2014-12-08)

Status: **Current**

Page 1 of 4

Date of Issue: **2018-02-28**

Applicant: **HKC Co., Ltd.**
26, Emtibeui 28-ro, Siheung-si,
Gyeonggi-do
Korea, Republic of

Equipment: **Limit switch box type APL-5*****

Optional accessory:

Type of Protection: **flameproof enclosure, intrinsic safety, dust ignition protection by enclosure**

Marking:

Ex d IIC T6 Gb
Ex ia IIC T6 Gb
Ex tb III C T85°C Db

Approved for issue on behalf of the IECEx
Certification Body:

Holger Schaffer

Position:

Certification Manager

Signature:
(for printed version)

Date:

2018-02-28



1. This certificate and schedule may only be reproduced in full.
2. This certificate is not transferable and remains the property of the issuing body.
3. The Status and authenticity of this certificate may be verified by visiting the [Official IECEx Website](http://www.iecex.com).

Certificate issued by:

Bureau Veritas Consumer Products Services Germany GmbH
Businesspark A96
86842 Türkheim
Germany





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Certificate No: IECEx EPS 14.0094X

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Page 2 of 4

Manufacturer: **HKC Co., Ltd.**
26, Emtibeui 28-ro, Siheung-si,
Gyeonggi-do
Korea, Republic of

Additional Manufacturing location(s):

This certificate is issued as verification that a sample(s), representative of production, was assessed and tested and found to comply with the IEC Standard list below and that the manufacturer's quality system, relating to the Ex products covered by this certificate, was assessed and found to comply with the IECEx Quality system requirements. This certificate is granted subject to the conditions as set out in IECEx Scheme Rules, IECEx 02 and Operational Documents as amended.

STANDARDS:

The electrical apparatus and any acceptable variations to it specified in the schedule of this certificate and the identified documents, was found to comply with the following standards:

IEC 60079-0 : 2011 Edition:6.0	Explosive atmospheres - Part 0: General requirements
IEC 60079-1 : 2007-04 Edition:6	Explosive atmospheres - Part 1: Equipment protection by flameproof enclosures "d"
IEC 60079-11 : 2011 Edition:6.0	Explosive atmospheres - Part 11: Equipment protection by intrinsic safety "i"
IEC 60079-31 : 2008 Edition:1	Explosive atmospheres – Part 31: Equipment dust ignition protection by enclosure "t"

*This Certificate **does not** indicate compliance with electrical safety and performance requirements other than those expressly included in the Standards listed above.*

TEST & ASSESSMENT REPORTS:

A sample(s) of the equipment listed has successfully met the examination and test requirements as recorded in

Test Report:

[DE/EPS/ExTR14.0095/01](#)

Quality Assessment Report:

[NL/DEK/QAR12.0070/03](#)



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Certificate No: IECEx EPS 14.0094X

Issue No: 1

Date of Issue: 2018-02-28

Page 3 of 4

Schedule

EQUIPMENT:

Equipment and systems covered by this certificate are as follows:

The series APL-5*** Limit Switch Box is an equipment which monitors the present position of a valve by contacting micro switches inside an Ex-d enclosure.

The enclosure is rated IP 66/67.

The interior of the enclosure is fitted with up to four switches rated at up to 16A, 250Vac with associated switch assembly and terminal block.

Option:

The limit switch box can be equipped with Ex ia proximity sensor NJ2-V3-N (Pepperl + Fuchs). The electrical connection must be made according to the following max. ratings and internal reactances:

$U_i = 16\text{ V}$ $I_i = 25\text{ mA}$ $P_i = 64\text{ mW}$ $C_i = 40\text{ nF}$ $L_i = 50\text{ }\mu\text{H}$

SPECIFIC CONDITIONS OF USE: YES as shown below:

Ambient temperature range: $-20^{\circ}\text{C} \leq T_a \leq +60^{\circ}\text{C}$

Cable entry holes are provided as specified on the certified drawings for the fitting of suitable certified flameproof cable entry devices. Unused entries are to be fitted with suitable certified flameproof blanking plugs.

For use with proximity sensor NJ2-V3-N the following additional condition applies:

The equipment shall only be connected to certified intrinsically safe evaluation and supply circuit with type of protection Ex [ia] IIC.



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Certificate No: IECEx EPS 14.0094X

Issue No: 1

Date of Issue: 2018-02-28

Page 4 of 4

DETAILS OF CERTIFICATE CHANGES (for issues 1 and above):

Issue 1:

New micro switch as internal component, modification of equipment description, change of manufacturer address